Faculty Placement Program

Post Placement Report

1. Name: Dr. Momina Moetesum
2. Designation: Head of Department, Software Engineering
3. Industry Visited: vResolv.io (https://vresolv.io/)
4. Location of the Industry: Head Office-Gulberg Greens Islamabad
5. Dates of Visit: 29th, 30th, & 31st July, 2025
6. Name of industry contact person: Mr. Aadil Jaleel Choudhry (Co-Founder)
7. Contact information: +92-321-5852-714

8. Introduction to the Industry:

vResolv.io is an innovation-driven technology company offering end-to-end solutions in Generative AI, Software Development, and Embedded Systems. Their expertise lies in MVP development, scalable product design, and staff augmentation. With a strong emphasis on cross-functional engineering, they cater to startups and enterprises in need of agile, modular tech solutions.

Visit Plan:

Date	Place/ Department	Comments
29th July 2025	Initial Meeting & Presentations	Introductory session with co-founders Mr. Daniyal Yasin and Mr. Aadil Jaleel. NUST and vResolv.io presented their capabilities and potential collaboration areas.
30th July 2025	AI Lab	Detailed walkthrough and discussions with AI engineers, including alumni from SEECS and FAST who are currently employed at vResolve.io.
30th July 2025	Embedded Systems Lab	Observed real-time embedded solutions, interacted with developers working on PCB design. Some were alumni from SMME and GIKI.
31st July 2025	Management Office/Final Debrief & Future Collaboration Discussion	Discussed potential collaboration on projects, internships, and training programs.

9. Details of the departmental visits

Day 1: 29th July 2025 – Initial Meetings & Presentations

Location: vResolv.io Management Office, Gulberg Greens, Islamabad

Key Personnel Met:

- Mr. Daniyal Yasin (Co-founder & CTO)
- Mr. Aadil Jaleel Choudhry (Co-founder & CEO)

Activities:

- Introductory session and formal welcome by the vResolv.io leadership.
- Overview presentation of NUST-SEECS covering faculty expertise, ongoing research, student projects, and collaboration models.
- Company presentation by vResolv.io, highlighting:
    - Key verticals including Generative AI, Embedded Systems, and Software Development.
    - Client portfolios, product development cycles, and innovation roadmap.
- Discussion of mutual interest areas including AI-based retail analytics, embedded solutions for energy optimization, and academic collaborations.
- Preliminary discussion on potential research and FYP collaborations.





Day 2: 30th July 2025 – Departmental Visits: AI & Embedded Systems Labs

Part 1: AI Lab Visit

Personnel Engaged:

- AI Engineers
- UI/UX and Design Engineers
- C++, Python, Full-Stack Developers

Activities:

- Demonstration of AI-powered tools like:
  - NVIDIA Jetson-Powered License Plate Insight System
  - Aquatic Object Detection & Tracking Ecosystem
  - AudioProof: Audio-to-Text Verification System
  - Omni-Directional Treadmill: VR Motion Tracking
- Discussion on:
  - Challenges in deploying real-time AI models on constrained hardware.
  - Performance optimization in real-time inference-based systems.

Key Observations:

- Emphasis on building cost-effective, edge-deployable AI solutions.
- Potential collaboration on Computer Vision and LLM-based projects.

Part 2: Embedded Systems Lab Visit

Personnel Engaged:

- Mechanical Engineers
- Developers working on IoT
- PCB Design Experts

Activities:

- Demonstration of:
  - vSense.cloud: Enterprise IoT Management Platform
  - Embedded vision modules used for drones and autonomous navigation.
  - Laser Profile Scanner for Welding Quality Control
  - Camera sensor calibration
  - Magnetic field visualization
- Discussion on:
  - Challenges in PCB prototyping, debugging, and real-time sensor calibration.
  - Interfacing AI models with microcontrollers using lightweight frameworks like TinyML and TensorFlow Lite.

Key Observations:

- Potential scope for joint research on hardware-software co-design.

Day 3: 31st July 2025 – Final Debrief & Proposal Discussions

Location: Management Office

Activities:

- Final debrief with co-founders and team leads to consolidate observations.
- Identification of potential areas where collaboration can be done.
- **Discussion on:**
  - Potential industry-based FYPs and research projects.
  - Co-organization of training programs and joint events like hackathons or coding competitions.
  - Requirement of specific expertise from NUST for consultancy or faculty placement.
  - Student internship and industrial visit facilitation.

Brainstorming possible funding avenues:

- Funding upto 100k for FYP (conditional to their interests, team and approvals)
- Joint proposal for external R&D grants (e.g., Ignite, HEC-NRPU).

Outcome:

- Consensus on beginning with a small-scale FYP + research pilot model.

- Exploration of opportunities for vResolv.io to sponsor or co-supervise 1–2 projects.
- Support of Student Industrial Visits.
- Further faculty placements and potential consultancy in Embedded System Design and Aero space Design.
- Potential joint trainings and events.

10. Needs and Problems faced by the industry related to your area of expertise

- Industry highlighted the need for fast, lightweight computer-vision models that can work in real time on resource-constrained platforms, an area that aligns with my work as a computer vision researcher.

- They are particularly interested in techniques like pruning, quantisation, and architecture simplification which relate to my experience in making CV models efficient for practical deployment.

- They also stressed on explaining the mathematical and theoretical justification of each technique, which matches my focus on my area of interest i.e. explainable AI.

11. Research / Industrial / Consultancy Projects Proposed:

Some research projects suggested by the industry are as follows:

S. No.	Title of the proposed research project
01	AI-Integrated Swarm Drones for Remote Navigation and Monitoring
02	Terrain-Following Autonomous Drone Navigation Using Vision-Based FP Control

The industry suggested that they can sponsor upto 100K for projects but conditional to their interest and team.

12. Trainings/Workshops Proposed:

The industry wants to organize collaborative trainings on the following topics.

S. No.	Title of training / workshop
01	Project management training
02	Generative AI training
03	Design sprints & rapid prototyping workshop

13. Internships/Jobs slots:

Currently, the industry does not want to provide temporary internship opportunities as they work on projects with NDAs with clients. However, I believe that a successful FYP and consultancy collaboration can pave way for future collaborative models.

I have seen several employees that are alumni of NUST from SEECS and SMME at the industry and currently they require following:

No. of Internship / Job Positions	Domain (Engineering, management etc)
1x Job	AI Engineer
1x Job	MERN Stack Engineer

14. Final Year Projects proposed for NUST Students:

No. of FYPs	**Domain**
04	**Engineering/AI**
Virtual Try-On System for Clothing Fit Visualization Using Web and Local Applications	
AI-Powered Edge Surveillance for Real-Time Shoplifting and Vandalism Detection	
Smart Examination Monitoring System Using Voice and Video Analytics to Detect Cheating	
IoT-Based Hybrid Smart Geyser with Automatic Ignition and Energy Optimization	

15. General Observations:

During my interaction with several industries, I have observed that despite having several collaborative opportunities, most shy away from giving internship or projects to students due to their NDAs with clients. Some are reluctant due to propriety rights issues.

16. Recommendations:

- Offer flexible IP models (company-retained or shared IP) to increase comfort levels.
- Create a secure industry sandbox collaboration environment hosted by NUST.
- Assign faculty as official project custodians to act as gatekeepers between students and industry.
- Start with low-risk, short PoC-style projects to build trust gradually.
- Embed industry-sourced problems into course projects under academic supervision.